# Know When to Fold: Distribution-Shift-Aware Selective Prediction for Protein–Ligand Co-Folding

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## Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, but a correlation is not a decision rule, and the correlation is weakest exactly where drug discovery operates: on novel pockets and novel chemotypes. We recast co-folding reliability as **selective prediction** and build a model-agnostic, training-free layer that turns confidence into a risk-controlled accept/abstain decision with a finite-sample guarantee. On the released Runs N' Poses benchmark (13,536 delivered poses, six co-folding models) we show three things. (1) A conformal selective-risk gate is valid on i.i.d. data. (2) That guarantee **breaks** under the novelty shift central to drug discovery: a gate whose marginal error looks compliant under-controls error 2–3× on novel-ligand strata, and calibrating on familiar ligands then deploying on novel ones violates the guarantee in 100% of runs (realized error 0.55 against a 0.20 target). (3) The break is repaired by group-conditional and weighted conformal keyed on training-set similarity, and a combined reliability score cuts the area under the risk-coverage curve (AURC) by roughly a quarter to two-fifths over native confidence (paired data-bootstrap over test poses excludes zero for every model), roughly doubling to tripling the predictions retained at a fixed guarantee. The layer is released as a pip-installable package that wraps frozen model outputs.

## 1. Introduction

Co-folding models are increasingly used to prioritise molecules and generate starting structures for downstream design. Recent work establishes that their confidence tracks accuracy and that accuracy drops on inputs dissimilar to training (Runs N' Poses, FoldBench). Neither gives a practitioner what they need: a rule for **which predictions to trust**, with a guarantee that survives the distribution shift they actually face.

We provide that rule. Our contribution is (i) selective prediction / conformal risk control for co-folding **pose** reliability, a structured 3D output rather than a scalar property. The closest prior work, CoDrug, conformalizes a scalar molecular property under shift, and we are not aware of a prior conformal treatment of pose correctness. Further, (ii) a sharp, quantified demonstration that standard conformal guarantees collapse under novel-pocket / novel-chemotype shift, using training-set structural and chemical similarity as the shift variable; and (iii) a shift-robust repair plus a combined reliability score, packaged as a training-free layer over frozen AlphaFold3 (AF3) / Boltz / Chai outputs.

## 2. Method

**Setup.** The unit is the delivered pose: the top-1 sample per (system, ligand, model) by the model's own ranking score. The label is Y = 1 iff the symmetry-corrected ligand root-mean-square deviation (RMSD) is $\leq$ 2 Å. The gate accepts a pose when its confidence s $\geq \tau$ and abstains otherwise; we report selective risk (error among accepted) against coverage (fraction accepted).

**Certified threshold.** We choose $\tau$ by Learn-then-Test with fixed-sequence testing [Angelopoulos et al. 2021]. Over a pre-specified coverage grid, each null hypothesis H0($\tau$): P(error | s $\geq \tau$) $\geq \alpha$ is tested with an exact binomial p-value P(Bin(n_accept, $\alpha$) $\leq$ errors); the accept set grows while H0 is rejected at level $\delta$ and stops at the first failure. This gives P(selective risk $\leq \alpha$) $\geq$ 1 - $\delta$, finite-sample and distribution-free. The binomial test conditions on the accept count, the correct tool for error-among-accepted.

**Novelty and shift-robustness.** Training-set similarity (ECFP4 Tanimoto to the nearest training ligand, pocket similarity, temporal cutoff), shipped pre-computed by Runs N' Poses, is the covariate-shift variable. It both stratifies (group-conditional / Mondrian conformal: a separate $\tau$ per novelty stratum) and weights (weighted conformal [Tibshirani et al. 2019]: reweight familiar-source calibration points by an estimated likelihood ratio to certify on a novel target without target labels). Ligands with no training analog form their own extreme stratum.

**Combined score.** Native ranking score is a weak sorter for some models. We fit a calibration-only combiner (gradient boosting $\rightarrow$ P(correct)) over native confidence, interface chain-pair ipTM, PoseBusters physical validity, intra-model ensemble spread across diffusion samples, and ligand difficulty. Novelty is excluded from the score and enters only through calibration. A 3-way split (fit combiner / calibrate $\tau$ / test) preserves conformal validity for the learned score: the combiner never sees the test fold and the threshold is calibrated on out-of-sample combiner scores, so there is no train/test leakage.

## 3. Results

Data: released Runs N' Poses predictions consumed as-is (no inference), 13,536 delivered poses, six models. $\alpha = 0.20$, $\delta = 0.10$ unless noted.

**Raw novelty gradient.** AF3 delivered-pose correctness falls from 0.88 on familiar ligands to 0.44 on the most novel with-analog stratum. A single global threshold cannot hold a uniform guarantee across this gradient.

**E1: valid i.i.d.** The certifier's finite-sample guarantee — true selective risk $\leq \alpha$ with probability 1 - $\delta$ — is validated on synthetic data where the true risk is known (the certified gate holds in $\geq$ 1 - $\delta$ of draws; the certifier is tight, so realized risk sits at $\alpha$ and a finite-test indicator is a noisy proxy). On RNP the certified gate's mean realized risk is $\leq \alpha$ for models whose native score certifies non-trivial coverage (AF3 0.17 at 28% coverage; Boltz similar). Native ranking score certifies almost nothing for Chai and Protenix (< 5% coverage): a near-vacuous gate that motivates the combined score below.

**E2: the break.** AF3's marginal risk (0.177) hides severe per-stratum under-control:

| novelty stratum | S0 | S1 | S2 | S3 | S4 |
|---|---|---|---|---|---|
| realized selective risk | 0.07 | 0.15 | 0.16 | **0.38** | **0.43** |

Calibrating on familiar ligands and deploying on novel ones accepts 98% of poses at realized error **0.55** with the guarantee holding in **0%** of runs (violated in 100%). All six models show the same collapse. The break is a property of structural/chemical novelty, not recency: it is at least as strong along pocket-novelty (AF3 S0 0.06 $\rightarrow$ S3 0.40, Chai up to 0.77) but weak along a temporal release-date axis, which sharpens what the shift variable must capture.

**E3: repair.** Group-conditional calibration (a separate $\tau$ per novelty stratum) restores the guarantee: no stratum's accepted error exceeds $\alpha$. With native confidence the cost is heavy abstention on the hardest strata (the gate folds rather than assure falsely). With the combined score, group-conditional calibration instead recovers usable coverage across the gradient while holding risk $\leq \alpha$: AF3 accepts 52% of stratum S1 and 39% of S2 (vs 8% and 12% with native confidence) and certifies a small fraction of the novel S3 that native confidence must abstain on. Only the no-training-analog extreme (S4) stays uncertifiable, where abstention is the correct answer.

**E4: utility.** The combined score dominates native confidence on the risk-coverage curve:

| model | AURC native | AURC combined | Δ |
|---|---|---|---|
| AF3 | 0.187 | 0.125 | -33.1% |
| Boltz-1 | 0.233 | 0.169 | -27.5% |
| Boltz-1x | 0.215 | 0.163 | -24.1% |
| Chai-1 | 0.246 | 0.149 | -39.5% |
| Protenix | 0.280 | 0.174 | -37.6% |

The AURC reduction is significant per model: a paired bootstrap over test poses gives 90% CIs on $\Delta$(AURC) that exclude zero (AF3 0.070 [0.049, 0.092], Protenix 0.092 [0.068, 0.116]; Boltz-2 omitted for power, n = 933). At $\alpha = 0.2$ the combined gate retains far more predictions at the same guarantee (AF3 coverage 0.22 $\rightarrow$ 0.71; Chai 0.00 $\rightarrow$ 0.61) and unlocks the stringent $\alpha = 0.1$ (90%-correct) operating point that native confidence cannot certify at all for several models. The combined score fuses native confidence with interface ipTM, PoseBusters validity, intra-model ensemble spread, and cross-model agreement.

**E3b: weighted repair, label-free.** Calibrating on familiar ligands and correcting with likelihood-ratio weights over the novelty covariates pulls the realized error on a moderately-novel target toward $\alpha$ without target labels (AF3 naive 0.27 $\rightarrow$ weighted 0.20 at 0.71 coverage; same trend for all models). This is a reweighted point estimate, not a finite-sample-certified gate; group-conditional calibration remains the rigorous fallback where stratum labels exist. On the extreme novel regime ($<$ 55% correct at baseline) weighted conformal reduces error (0.55 $\rightarrow$ 0.31) but no gate can certify a high-coverage low-error set, so the layer correctly abstains.

**E6: downstream payoff.** The gate cleans the delivered pose set a downstream pipeline would carry forward: base top-1 purity 63–70% rises to 82% ($\alpha$=0.2, retaining 81% of correct AF3 poses) or 93–94% ($\alpha$=0.1, all models), with interface LDDT-PLI lifting from ~0.72 to 0.85–0.92. Downstream structure-based work receives near-clean inputs instead of a one-in-three error rate. The Mac1 virtual-screen enrichment arm follows when its coordinates release.

**E5: robustness and ablation.** The break and the gain do not depend on the 2 Å convention: across thresholds 1.5–3.0 Å the novel strata always exceed $\alpha$ (AF3 S3 risk 0.35–0.43 vs S0 0.05–0.09) and the combined score cuts AURC 26–29%. A cumulative feature ablation attributes the gain mainly to interface ipTM (AURC 0.19 $\rightarrow$ 0.16), intra-model ensemble spread (0.16 $\rightarrow$ 0.14), and cross-model agreement (0.14 $\rightarrow$ 0.13); PoseBusters and ligand difficulty add little on top.

**E8/E9: task- and label-agnostic.** The layer is not tied to the 2 Å pose label. On interface quality (local distance difference test on the protein-ligand interface, LDDT-PLI $\geq$ 0.5) the combined score again dominates native confidence on AURC (44–55% lower). Dropping the threshold entirely and ordering by continuous RMSD, the accepted set's mean RMSD at 50% coverage falls from ~2.1 Å (native) to ~1.1–1.5 Å (combined).

## 4. Discussion and limitations

The guarantee is over the chosen correctness definition and the sampled shift axes, not a universal notion of trust; the findings are robust across RMSD thresholds 1.5–3.0 Å (E5), though a continuous-RMSD risk is left for future work. Weighted conformal coverage is approximate under estimated weights, so we keep group-conditional as the rigorous fallback. A cross-dataset test on FoldBench did not replicate the advantage: its public per-pose table ships only ranking_score, without the interface-ipTM / PoseBusters / ensemble features the ablation identifies as the source of the gain, so the feature-poor combiner does not beat raw confidence there. A clean second-benchmark test needs a release that exposes those confidences per pose. On the extreme novel-chemotype regime (base correctness $<$ 55%) no gate can certify a high-coverage low-error set; the layer's value there is principled abstention over a naive gate's false confidence. Code, calibration tables, and a one-command reproduction pipeline are provided in the accompanying repository (public release on acceptance).

## References

Abbreviated; full entries in the repository `REFERENCES.bib`. Angelopoulos, Bates, Candès, Jordan, Lei. Learn then Test. 2021. Bates, Angelopoulos, Lei, Malik, Jordan. Risk-Controlling Prediction Sets. JACM 2021. Tibshirani, Foygel Barber, Candès, Ramdas. Conformal Prediction Under Covariate Shift. NeurIPS 2019. Škrinjar et al. Have protein–ligand co-folding methods moved beyond memorisation? (Runs N' Poses) 2025. Buttenschoen, Morris, Deane. PoseBusters. Chem. Sci. 2024.